

3

Gravitation



Gravitational force holds artificial satellites in their orbit around the earth. They are launched from earth for purposes of scientific research, communication, weather forecasting, intelligence, etc.

“Gravitation is a force of attraction that acts between all objects in the universe.”

1. Introduction

Gravitation is the weakest force in nature. It is negligible in the interactions of tiny particles, and thus plays no role in molecules, atoms, and nuclei. The gravitational attraction between objects of ordinary size, such as the gravitational force exerted by a building on a car, is too small to be noticed. When we consider very large objects, such as stars, planets, and satellites (moons), gravitation is of primary importance. The gravitational force exerted by the Earth on us and on the objects around us is a fundamental part of our experience. It is gravitation that binds us to the Earth and keeps the Earth and the other planets on course within the solar system. The gravitational force plays an important role in the life history of stars and in the behavior of galaxies.

2. The Newtonian gravitation

Sir Isaac Newton did not discover gravitation, its effects have been known throughout human existence. But he was the first one to understand the broader significance of gravitation. Newton discovered that ‘gravitation is universal, it is not restricted to Earth only’, as others physicists of his time assumed.

According to popular legend, Newton was sitting under an apple tree, when an apple landed on his head. He looked up at the sky, noticed the Moon, and reasoned that the same force that made the apple fall to the ground also kept the Moon in its orbit about the Earth.



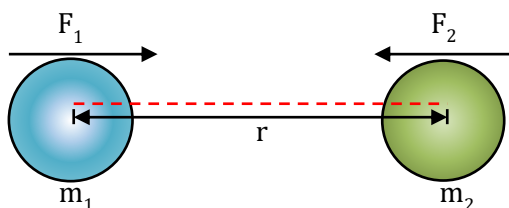
The legend ‘apple falling on Newton’s head’.

The fact that the Moon did not move in a straight line suggested to him that there must be a force acting on it. Newton gave an idea that no-one else had even considered that 'gravity from the Earth acted through space and exerted a force on the Moon'.

3. Newton's universal law of gravitation

'Every object in the universe attracts every other object with a force which is proportional to the product of their masses and inversely proportional to the square of the distance between them. The force is along the line joining the centers of the two objects'.

Let us consider two masses m_1 and m_2 lying at a separation distance r . Let the force of attraction between the two objects be F .



Newton's universal law of gravitation

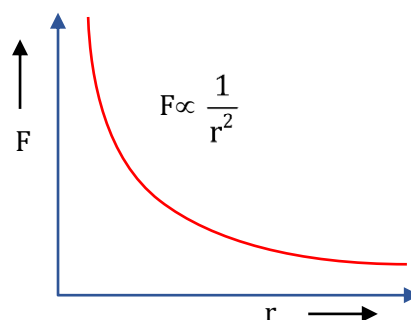
According to the universal law of gravitation, $F \propto m_1 m_2$

Also, $F \propto \frac{1}{r^2}$ (see figure)

Combining both, we get, $F \propto \frac{m_1 m_2}{r^2}$

or
$$F = \frac{G m_1 m_2}{r^2}$$

where, G is the constant of proportionality and it is called the 'universal gravitational constant'.



Dependence of gravitational force (F) on separation distance (r)

Definition of universal gravitational constant

Universal gravitation constant is the magnitude of the force (in newton) between a pair of 1 kg masses that are kept 1 meter apart.

Unit of universal gravitational constant

According to the universal law of gravitation,

$$F = \frac{G m_1 m_2}{r^2} \quad \text{or} \quad G = \frac{F r^2}{m_1 m_2}$$

$$\text{Thus, SI unit of } G = \frac{\text{Newton} \times (\text{meter})^2}{\text{kg} \times \text{kg}} = \frac{\text{Nm}^2}{\text{kg}^2} = \text{Nm}^2 \text{kg}^{-2}$$

☞ The numerical value of G is $6.673 \times 10^{-11} \text{ N m}^2 \text{kg}^{-2}$.

4. Characteristics of gravitational force

- (1) It is a universal force of attraction.
- (2) It acts along the line joining the center of each mass.
- (3) It acts equally on each mass, i.e., it obeys Newton's third law. $F_{12} = -F_{21}$
- (4) It is weaker if the masses are further apart. Gravitation acts in an inverse square manner, i.e. $F \propto \frac{1}{r^2}$, where r is the distance between the centers of the masses.
- (5) It depends directly on the mass of each body involved, i.e. $F \propto m_1$ and $F \propto m_2$.
- (6) It is a long-range force i.e., its influence extends to very large distances.

Gravitational force between two masses does not depend on the medium present between them. Gravitational force between two masses does not depend on the presence of other masses around them.



Do You Remember ?

- ★ Gravitation is the weakest fundamental force that exist in nature.



Building Concepts

1

Two objects A and B of masses m_1 and m_2 respectively are kept r distance apart (given, $m_1 > m_2$). (a) Which object will exert more gravitational force on the other? (b) If distance between them is made $6r$, how does the gravitational force between them change? (c) If an object C of mass m_3 is placed between them, how does the gravitational force between object A and B change?

Explanation

- (a) Both will exert equal force on each other, they exist in an action-reaction pair.
- (b) If distance is made $6r$, the force between them becomes $\frac{1}{36}$ times the original value.

This is because gravitational force follows inverse square relationship with the separation distance r .

- (c) Gravitational force between objects A and B remains the same as it does not depend on the other masses present near or in-between them.

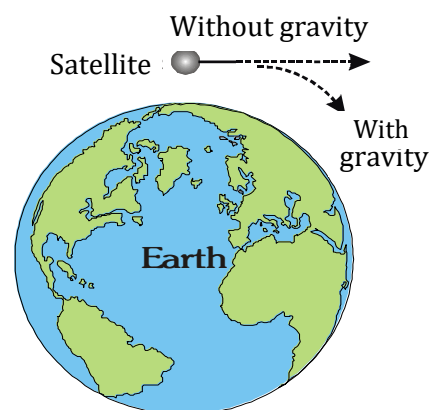
5. Importance of Newton's universal law of gravitation

Newton's universal law of gravitation successfully explained several phenomena which were believed to be unconnected:

- (1) The force that binds us to the Earth.
- (2) The motion of the Moon around the Earth.
- (3) The motion of planets around the Sun.
- (4) The tides due to the Moon and the Sun.

6. Why Moon does not fall on Earth directly?

The motion of Moon is just like the motion of an object in circular motion. The velocity of Moon is directed tangential to the circle at every point along its path. The acceleration of Moon is directed towards the center of the circle i.e., towards the Earth (the central body) around which it is orbiting. This acceleration is caused by a centripetal force which is supplied by the gravitational force between the Earth and the Moon. If this force were absent, the Moon would continue in motion at the same speed in a direction tangential to the circular path and would have escaped away from the Earth. If the Moon had no tangential velocity, it would have fallen on Earth due to gravitation. Thus, it is the tangential velocity and the gravitational force that are perpendicular to each other, and cause the Moon to fall around the Earth without actually falling onto it (see figure).



If gravitational force becomes zero, then a satellite (moon) will go away from Earth along a tangential path.



Check your
Concepts

1

1. When a person passes you on the street you do not feel a gravitational pull. Why?
2. Acceleration of an object on earth by applying the force 'F' is 'a'. To achieve the same acceleration on the Moon, will the net force acting on the object be greater than, less than, or the same as that required on earth?
3. One object has a mass m_1 and a second object has a mass m_2 , which is greater than m_1 . Does the mass m_2 exert more force on m_1 ?



Quick
Tips

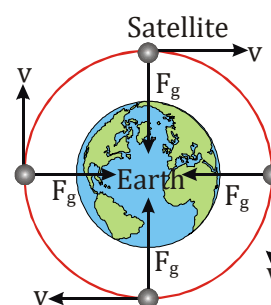
- ★ $a^m \times a^n = a^{m+n}$
- ★ $\frac{a^m}{a^n} = a^{m-n}$
- ★ $\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m$



Numerical
Ability

1

1. The mass of the Earth is 6×10^{24} kg and that of the Moon is 7.4×10^{22} kg. If the distance between the Earth and the Moon is 3.84×10^5 km, calculate the force exerted by the Earth on the Moon. ($G = 6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$)



The gravitational force acting on a satellite in a circular orbit is always perpendicular to the velocity of the satellite.

Solution

Mass of Earth, $M = 6 \times 10^{24}$ kg ; mass of Moon, $m = 7.4 \times 10^{22}$ kg, $G = 6.7 \times 10^{-11}$ N m² kg⁻²

Distance between Earth and Moon, $r = 3.84 \times 10^5$ km = $3.84 \times 10^5 \times 1000$ m = 3.84×10^8 m

From universal law of gravitation,

$$F = \frac{GMm}{r^2} = \frac{(6.7 \times 10^{-11})(6 \times 10^{24})(7.4 \times 10^{22})}{(3.84 \times 10^8)^2}$$

$$= \frac{6.7 \times 6 \times 7.4 \times 10^{24+22-11}}{3.84 \times 3.84 \times 10^{16}} = \frac{297.48 \times 10^{35-16}}{14.746}$$

$$= 20.1 \times 10^{19} = \mathbf{2.01 \times 10^{20} \text{ N}}$$

Decode the problem

To find the force we should know the m_1 , m_2 and r

Apply the formula

$$F = \frac{Gm_1m_2}{r^2}$$

2. A planet of mass M has two identical moons, each of mass m . Moon 1 is in a circular orbit of radius r and Moon 2 is in a circular orbit of radius $2r$. What is the ratio of magnitude of the gravitational force exerted by the planet on Moon 1 to the magnitude of gravitational force exerted by the planet on Moon 2?

Solution

Let the magnitudes of gravitational forces exerted by the planet on Moon 1 and Moon 2 are F_1 and F_2 respectively.

$$\text{Now, } F_1 = \frac{GMm}{r^2} \quad \dots (1)$$

$$\text{and, } F_2 = \frac{GMm}{(2r)^2} \text{ or } F_2 = \frac{GMm}{4r^2} \quad \dots (2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{F_1}{F_2} = \frac{\frac{GMm}{r^2}}{\frac{GMm}{4r^2}} = \frac{GMm}{r^2} \times \frac{4r^2}{GMm} = \frac{4}{1}$$

$$\therefore F_1 : F_2 = 4 : 1$$

3. The gravitational force between two objects is 49 N. How much must the distance between these objects be decreased so that the force between them becomes double?

Solution

Here, $F = 49$ N

Let r = original distance between the objects, m_1 and m_2 = masses of the objects

Gravitational force between these objects is given by

$$F = \frac{Gm_1m_2}{r^2} \quad \dots (1)$$

$$\text{Now, } F' = 2F = 2 \times 49 \text{ N} = 98 \text{ N}$$

Decode the problem

To find the force we should know the m_1 , m_2 and r

Apply the formula

$$F = \frac{Gm_1m_2}{r^2}$$

Decode the problem

To find the distance we should know the m_1 , m_2 and F .

Apply the formula

$$F = \frac{Gm_1m_2}{r^2}$$

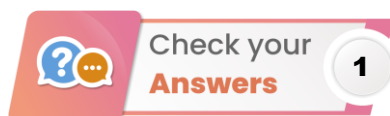
Let, r' = new distance between the objects, m_1 and m_2 = masses of the objects
Gravitational force between these objects is given by

$$F' = \frac{Gm_1m_2}{r'^2} \quad \dots (2)$$

Dividing (1) by (2), we get

$$\frac{F}{F'} = \frac{r'^2}{r^2} \quad \text{or} \quad \frac{r'}{r} = \sqrt{\frac{F}{F'}} = \sqrt{\frac{F}{2F}} = \frac{1}{\sqrt{2}} \quad [\square F' = 2F]$$

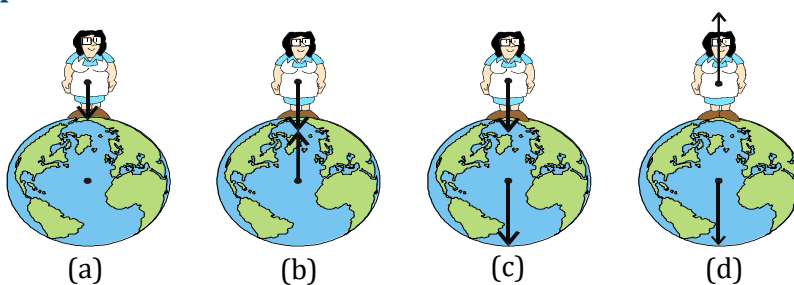
$$\therefore r' = \frac{r}{\sqrt{2}}$$



1. A person passing you on the street exerts a gravitational force on you, but it is so weak (about 10^{-7} N or less) that it is insignificant.
2. The net force required to accelerate the object is specified by Newton's second law as $F = ma$, where 'm' is the object's mass and 'a' is the acceleration along the ground. For a given acceleration, the net force depends only on the mass. But the mass of the object is the same on the Moon as it is on the Earth. Therefore, the same net force would be required for a given acceleration on the Moon as on the Earth.
3. No, both will exert equal forces on each other which are opposite in direction. This is in agreement with Newton's third law.



Which figure best represents the gravitational force acting on you and on Earth (Figure)? Explain.



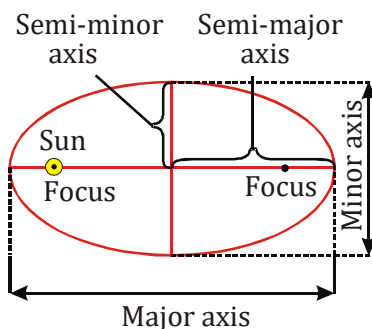
Building Concepts 2

Explanation

The gravitational forces exist in action-reaction pair. The force on you should be towards the centre of Earth (or vertically downward). The force on earth will be opposite to that acting on you i.e., it should be away from the centre of Earth (vertically upwards). Thus, [figure (b)] best represents the gravitational force acting on you and on Earth.

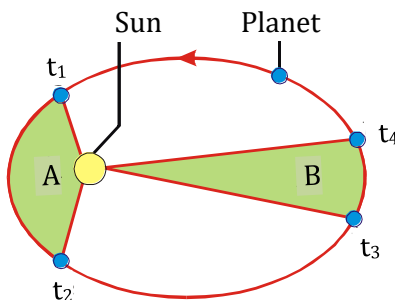
7. Kepler's laws of planetary motion

- Each planet in the Solar system moves in an elliptical orbit with the Sun at one focus (see figure).



Elliptical path of a planet

- The line joining the planet and the Sun sweeps out equal areas in equal intervals of time (see figure).



Kepler's Second law. For equal time intervals $(t_2 - t_1)$ and $(t_4 - t_3)$, Area A = Area B.

- The square of time period of a planet revolving about the Sun is proportional to the cube of the mean distance of the planet from the Sun.

In other words, 'the square of the orbital period of a planet is proportional to the cube of the semimajor axis of the elliptical orbit'.

That is, $T^2 \propto R^3$

or $\boxed{\frac{T^2}{R^3} = \text{Constant}}$



Let us consider two planets having orbital time periods T_1 and T_2 respectively. Let their mean distances from the Sun are R_1 and R_2 respectively.

Then,

or $\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$

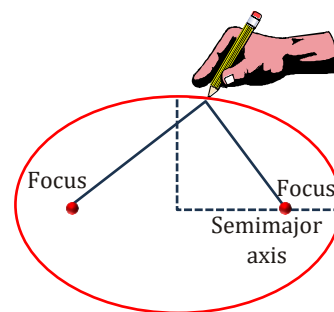
or $\boxed{\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{R_1}{R_2}\right)^3}$



According to Kepler's first law, planets follow elliptical orbits, with the Sun at one focus. How can you draw an ellipse?

Explanation

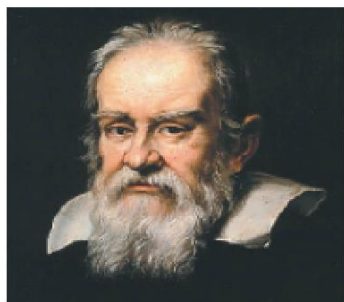
To draw an ellipse, fix two pins on a piece of paper fixed on a cardboard or a drawing board. Two pins represent the two foci of the ellipse. Now, connect a length of string to the two pins. Use a pencil and the string to sketch out a smooth, closed curve (see figure). This closed curve is an ellipse.



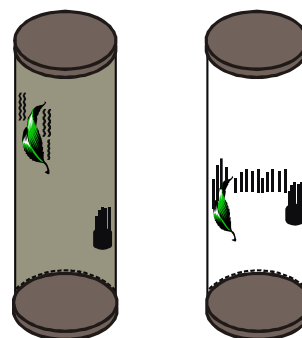
Active Physics 1

8. Free fall (motion under gravity)

Till 1600 AD, the teachings of the Greek philosopher Aristotle (384–322 BC) had held that *heavier objects fall faster than lighter ones*. The Italian physicist Galileo Galilei gave the present-day ideas of falling objects. Now, it is an established fact that, in the absence of air resistance, all objects dropped near the Earth's surface fall with the same constant acceleration under the influence of the Earth's gravity (see figure).



Galileo Galilei (1564–1642), Italian physicist and astronomer, formulated the laws that govern the motion of objects in free-fall.



Air filled tube Evacuated tube
In the absence of air, all objects fall with same acceleration (g).

- ☞ **Free fall** is the motion of an object subject only to the influence of gravity. An object is in free fall as soon as it is dropped from rest, thrown downward or thrown upward.

Acceleration due to gravity

The constant acceleration of a freely falling body is called the acceleration due to gravity.

- ☞ The acceleration due to gravity is the acceleration of an object in free fall that results from the influence of Earth's gravity. Its magnitude is denoted with the letter ' g '.

Acceleration due to gravity at the surface of Earth

Let us consider an object of mass m placed on the surface of Earth (see figure).

Let the mass of Earth be M and radius of Earth be R . The gravitational force on the object due to Earth is given by,

$$F_g = \frac{GMm}{R^2} \quad \dots(1)$$

Let this force produces an acceleration 'a' on the object, then,

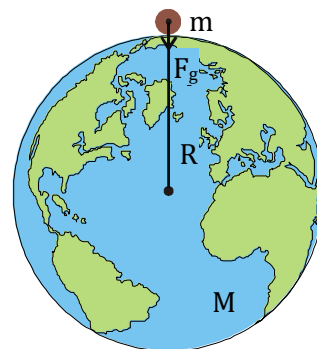
$$F_g = ma \quad \dots(2)$$

From eq.(1) and eq.(2), we get,

$$ma = \frac{GMm}{R^2} \quad \text{or} \quad a = \frac{GM}{R^2}$$

This acceleration is called acceleration due to gravity and it is denoted by g i.e., $a = g$

$$g = \frac{GM}{R^2}$$



Finding acceleration due to gravity on the Earth's surface

Calculating the value of g on Earth

Mass of the Earth = 6×10^{24} kg ; radius of Earth = 6.4×10^6 m ; $G = 6.673 \times 10^{-11}$ N m² kg⁻²

$$\text{Now, } g = \frac{GM}{R^2} = \frac{(6.67 \times 10^{-11})(6 \times 10^{24})}{(6.4 \times 10^6)^2} = 9.8 \text{ ms}^{-2}$$



The acceleration due to gravity g for any planet is

(i) directly proportional to the mass of the planet

(ii) inversely proportional to the square of the radius of the planet.



If the Earth were to suddenly shrink to half of its current radius with its mass remaining constant, would the acceleration due to gravity increase, decrease or stay the same?

Explanation

Acceleration due to gravity (g) is directly proportional to the mass of Earth and inversely proportional to the square of its radius. Here, mass of Earth is given constant, while radius is shrunk to half of its current radius. Thus, acceleration due to gravity will increase and in this case it will be four times of its original value.

9. Equations of motion of freely falling body

There are two main assumptions in free fall :

1. Acceleration due to gravity (g) is constant throughout the motion and it acts vertically downwards.
2. Air resistance is negligible.

Case 1 : An object thrown vertically upward and it returns after some time.

Let us consider an object (see figure) thrown vertically upward with an initial velocity u ; the acceleration due to gravity g is acting vertically downward on it. Let after a time interval t , it achieves a height h and final velocity v .

Initial velocity = $+u$; acceleration, $a = -g$; distance travelled, $s = +h$

From first equation of motion, we have, $v = u + at$

or $v = (+u) + (-g)t$

$$\text{or } \boxed{v = u - gt} \quad \dots(1)$$

From second equation of motion, we have, $s = ut + \frac{1}{2}at^2$

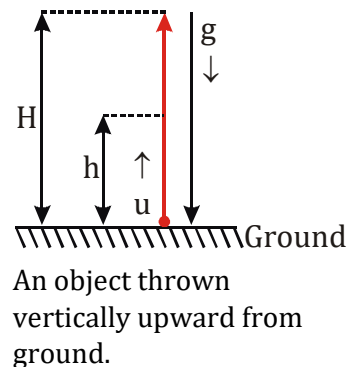
$$\text{or } +h = (+u)t + \frac{1}{2}(-g)t^2$$

$$\text{or } \boxed{h = ut - \frac{1}{2}gt^2} \quad \dots(2)$$

From third equation of motion, we have, $v^2 = u^2 + 2as$

$$\text{or } v^2 = (+u)^2 + 2(-g)(+h)$$

$$\text{or } \boxed{v^2 = u^2 - 2gh} \quad \dots(3)$$



(a) Time taken to reach maximum height

At maximum height, $v = 0$

From eq.(1), we get, $0 = u - gt$, or $u = gt$

$$\text{or } \boxed{t = \frac{u}{g}}$$

(b) Total time of journey

Since g is constant throughout the motion, time taken to reach maximum height from the ground is equal to time taken to reach ground from the maximum height. That is, total time (T) of journey,

$$T = 2t = \frac{2u}{g} \quad \text{or} \quad \boxed{T = \frac{2u}{g}}$$

(c) Maximum height achieved by the object

Let the maximum height achieved be H .

At maximum height, $v = 0$

From eq.(3), we get, $(0)^2 = u^2 - 2g(H)$

$$\text{or } u^2 = 2gH$$

$$\text{or } \boxed{H = \frac{u^2}{2g}}$$

☞ Here, the total distance covered, $s = 2H = 2\left(\frac{u^2}{2g}\right) = \frac{u^2}{g}$ while, the total displacement is zero.

Case 2 : An object is thrown vertically downward from a certain height H .

Let us consider an object (see figure) thrown vertically downward with an initial velocity u ; the acceleration due to gravity g is acting vertically downward on it. Let after a time interval t , it falls through a distance y and achieves a final velocity v .

Initial velocity = $-u$; acceleration, $a = -g$; displacement, $s = -y$;

final velocity = $-v$

From first equation of motion, we have, $v = u + at$

$$\text{or } (-v) = (-u) + (-g)t$$

$$\text{or } -v = -u - gt$$

$$\text{or } -v = -(u + gt)$$

$$\text{or } \boxed{v = u + gt} \quad \dots(1)$$

From second equation of motion, we have, $s = ut + \frac{1}{2}at^2$

$$\text{or } -y = (-u)t + \frac{1}{2}(-g)t^2$$

$$\text{or } -y = -ut - \frac{1}{2}gt^2$$

$$\text{or } -y = -\left(ut + \frac{1}{2}gt^2\right)$$

$$\text{or } \boxed{y = ut + \frac{1}{2}gt^2} \quad \dots(2)$$

From third equation of motion, we have, $v^2 = u^2 + 2as$

$$\text{or } (-v)^2 = (-u)^2 + 2(-g)(-y)$$

$$\text{or } \boxed{v^2 = u^2 + 2gy} \quad \dots(3)$$

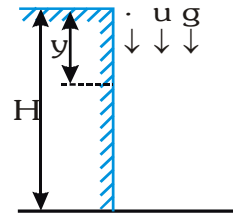
☞ For numerical, we can assume acceleration due to gravity as $+g$ for downward, while $-g$ for upward motion.

☞ If an object is dropped from certain height, its initial velocity is taken zero i.e., $u = 0$. In such case, the eq (1),(2),(3) will be reduced to,

$$v = gt ;$$

$$y = \frac{1}{2}gt^2 ;$$

$$v^2 = 2gy$$



An object thrown vertically downward from a certain height H .

**Be Alert !**

- ★ All objects, whether they are heavy or light, experience same value of acceleration due to gravity (g) i.e., g does not depend on the masses of individual objects.

**Quick****Tips**

- ★ We can also use the three equations of motions to solve the free fall problems.
- ★ We should take care of sign conventions to put the given value of equation.
- ★ To avoid mistake, take upper direction is positive & downward direction is negative.

**Numerical****Ability****2**

1. A body falls freely from rest. Find (a) the distance it falls in 3 seconds (b) its speed after falling 20 m (c) the time required to reach a speed of 25 m/s (d) time taken to fall 80 m. Take $g = 10 \text{ m/s}^2$.

Solution

Given, initial velocity, $u = 0$; $g = 10 \text{ m/s}^2$

(a) Distance travelled $s = ?$; time $t = 3 \text{ s}$

$$\text{Now, } s = ut + \frac{1}{2}gt^2$$

$$\text{or } s = (0)t + \frac{1}{2}(10)(3)^2 \quad \text{or } s = 45 \text{ m}$$

(b) Speed, $v = ?$; $s = 20 \text{ m}$

$$\text{Now, } v^2 = u^2 + 2gs = (0)^2 + 2(10)(20) = 400$$

$$\text{or } v = \sqrt{400} \quad \text{or } v = 20 \text{ m/s}$$

(c) Time $= ?$; $v = 25 \text{ m/s}$

$$\text{Now, } v = u + gt$$

$$\text{or } 25 = 0 + (10)t \quad \text{or } t = 2.5 \text{ s}$$

(d) Time $= ?$; distance travelled $s = 80 \text{ m}$

$$\text{Now, } s = ut + \frac{1}{2}gt^2$$

$$\text{or } 80 = (0)t + \frac{1}{2}(10)t^2 \quad \text{or } 80 = 5t^2$$

$$\text{or } t^2 = 16 \quad \text{or } t = \sqrt{16} \quad \text{or } t = 4 \text{ s}$$

Decode the problem

Identify the equation

$$s = ut + \frac{1}{2}at^2$$

(for part (i), since, u , a and t are given)

$$v^2 = u^2 + 2as$$

(for part (ii) since, u , a and s are given)

$$v = u + at$$

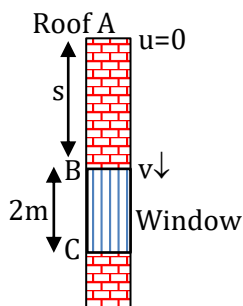
(for part (iii) since, u , a and v are given)

$$s = ut + \frac{1}{2}at^2 \quad (\text{for part (iv), since, } u, a \text{ and } s \text{ are given})$$

and s are given)

2. A boy drops a stone from the edge of the roof. It passes a window 2 m high in 0.1 s. How far is the roof above the top of the window? (Take $g = 10 \text{ ms}^{-2}$)

Solution



Numerical Ability 1(2)

Decode the problem

Identify the equation

$$s = ut + \frac{1}{2}at^2$$

(since, s, a and t are given)

$$v^2 = u^2 + 2as$$

(since, u, a and v are given)

Let a stone be dropped from the edge of the roof A. Let it passes over B with a velocity, say v. Consider motion BC.

$$u = v, a = 10 \text{ ms}^{-2}; s = 2\text{m}; t = 0.1 \text{ s}$$

Using, $s = ut + \frac{1}{2}gt^2$, we have

$$2 = v(0.1) + \frac{1}{2} \times 10 \times (0.1)^2$$

$$2 = 0.1v + 0.05 \Rightarrow 0.1v = 2 - 0.05 \quad \text{or} \quad v = \frac{1.95}{0.1} = 19.5 \text{ ms}^{-1}$$

This initial velocity at B in motion BC is the final velocity in motion AB. Considering motion AB, we have $u = 0$; $v = 19.5 \text{ ms}^{-1}$; $s = ?$; $a = 10 \text{ ms}^{-2}$

Using, $v^2 - u^2 = 2as$, we have

$$(19.5)^2 - (0)^2 = 2 \times 10 \times s$$

$$\text{or } s = \frac{19.5 \times 19.5}{2 \times 10} \text{ or } s = \mathbf{19.01 \text{ m}} \therefore \text{Roof is 19.01 m above the window.}$$

3. A ball thrown up is caught by the thrower after 4 s. With what velocity was it thrown up? How high did it go? Where was it after 3 s? ($g = 9.8 \text{ m s}^{-2}$)

Solution

Since the time of going up is the same as that of coming down, therefore time of going up = $4/2 = 2 \text{ s}$. Let it starts upward

$$\text{or } 0 = u - 9.8 \times 2$$

$$\text{or } \mathbf{u = 19.6 \text{ m s}^{-1}}$$

$$\text{Again } v^2 - u^2 = 2as$$

$$0 - (19.6)^2 = 2(-9.8)h$$

$$\text{or } \mathbf{h = 19.6 \text{ m}}$$

Decode the problem

Identify the equation

$$v = u + at$$

(for part (i) since, t, a and v are given)

$$v^2 = u^2 + 2as$$

(for part (ii) since, u, a and v are given)

$$s = ut + \frac{1}{2}at^2$$

(for part (iii), since, u, a and t are given)

After 2 s, it starts coming downwards (starting with $u = 0$). Considering downward motion,
 $u = 0$; $a = 9.8 \text{ m s}^{-2}$; $t = 3 - 2 = 1 \text{ s}$; $s = ?$

$$s = ut + \frac{1}{2} at^2 \quad \text{or} \quad s = 0 + \frac{1}{2} \times 9.8 (1)^2$$

= 4.9 m from top.

4. Coconut is hanging on a tree at a height of 15 m from the ground. A boy launches a projectile vertically upwards with a velocity of 20 m s^{-1} . After what time the projectile will pass by the coconut? Explain the two answers in this problem.

Solution

Here $u = 20 \text{ m s}^{-1}$; $a = -10 \text{ m s}^{-2}$; $s = 15 \text{ m}$; $t = ?$

Using $s = ut + \frac{1}{2} at^2$, we have

$$15 = 20t + (-10) t^2 \Rightarrow 15 = 20t - 5t^2$$

$$\text{or } 3 = 4t - t^2$$

$$\text{or } t^2 - 4t + 3 = 0$$

$$\text{or } (t - 1) (t - 3) = 0$$

$$\therefore t - 1 = 0 \quad \text{or} \quad t = 1 \text{ s}$$

$$t - 3 = 0$$

$$\text{or } t = 3 \text{ s}$$

After 1 s, it will cross coconut while going up and after 3 s, while coming down.

- ☞ While a ball is in free-fall, does its acceleration increase, decrease, or remain constant ?
 In free fall, acceleration due to gravity remains constant. It is velocity which decreases during upward motion and increases during downwards motion.

10. Mass

The amount of matter contained in a body is called its **mass**.

- ☞ Mass of a body is constant at all places in the universe. It is measured by physical balance. It is a scalar quantity and it is always taken positive.

Unit of mass

SI unit : Kilogram (kg) ; CGS unit : Gram (g)

11. Weight

The gravitational force exerted by the Earth on an object is called its weight. This force is directed towards the center of the Earth. It is denoted by letter W.

- ☞ The force with which an object is attracted towards the centre of Earth is called its **weight**. It is measured by a weighing machine or spring balance.

$$\text{Weight, } W = mg$$

Where, m = mass of object ; g = acceleration due to gravity

Unit of weight

SI unit : Newton (N) ; CGS unit : Dyne

**Be Alert !**

★ Mass and weight are different physical quantities.

Variation of weight of an object

Since weight depends on g , weight varies with location. Objects weigh less at higher altitudes than at sea level, because g decreases with increasing distance from the center of the Earth. Hence, weight, unlike mass, is not an inherent property of an object.

☞ Mass and weight are two different quantities. Mass of a body remains constant at all places, while weight of a body may change from place to place. Weight of a freely falling body is zero. At the centre of Earth, weight of a body is zero.

Weightlessness

We sense weight when something, such as the floor or chair, exerts a contact force on us. But if our chair and the floor, all are accelerating towards the Earth together like free fall, then no contact forces are exerted on us. Thus, our apparent weight is zero and we experience weightlessness.

The space shuttle orbits nearly 400 km above Earth's surface. At this distance, $g = 8.7 \text{ m/s}^2$, only slightly less than on Earth's surface. This means Earth's gravitational force is certainly not zero in the space shuttle. But, the astronauts experience weightlessness as the shuttle and everything in it falls freely towards the Earth.

☞ If a man is standing in a lift and the lift cable breaks, then the lift and the man will fall towards Earth with an acceleration g . The man is now in free fall, his apparent weight is zero and he will experience weightlessness.



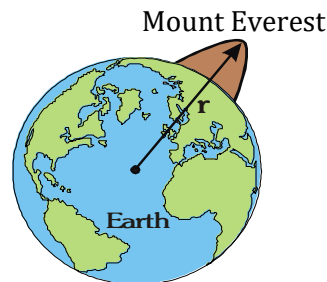
Acceleration due to gravity (g) is maximum at the surface of Earth. As we go above or below the surface of Earth, value of g decreases. At the centre of Earth, value of g is zero.

SPOT LIGHT
**Building
Concepts**
4

(i) If you climb to the top of Mt. Everest, how does the acceleration due to gravity change as compared to that at sea level? (ii) Is there any change in your weight? (iii) Is there any change in your mass?

Explanation

(i) As you climb to the top of Mt. Everest, distance (r) between you and the center of earth increases. As a result, the acceleration due to gravity (g) decreases, as it is inversely proportional to square of the distance (r) (see figure).



Building Concepts 4

- (ii) Now, weight (W) of an object is directly proportional to acceleration due to gravity (g). Since, value of g decreases, your weight also decreases.
- (iii) Your mass will remain constant as it does not depend on g . Mass is simply the amount of matter contained in an object.

Weight of an object on Moon

We know that the weight of an object on the Earth is the force with which the earth attracts the object. Similarly, the weight of an object on the Moon is the force with which the Moon attracts that object.

Let us consider an object of mass m . Let its weight on the Moon be W_m . Let the mass of the Moon be M_m and its radius be R_m .

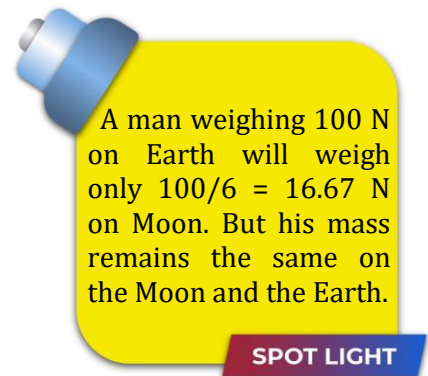
By Newton's universal law of gravitation, the weight of the object on the Moon will be,

$$W_m = \frac{GM_m m}{R_m^2} \quad \dots (1)$$

Let weight of the same object on the Earth be W_e . Let the mass of the Earth be M_e and its radius be R_e . Thus, the weight of the object on the Earth will be,

$$W_e = \frac{GM_e m}{R_e^2} \quad \dots (2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{W_m}{W_e} = \frac{\frac{GM_m m}{R_m^2}}{\frac{GM_e m}{R_e^2}} = \frac{M_m R_e^2}{M_e R_m^2} \quad \dots (3)$$



Now, mass of Earth is approximately 100 times the mass of Moon and radius of earth is nearly 4 times the radius of Moon i.e.,

$$M_e = 100 M_m ; \quad \dots (4)$$

Using (3) and (4), we get,

$$\frac{W_m}{W_e} = \frac{M_m (4R_m)^2}{100 M_m R_m^2} = \frac{M_m 16 R_m^2}{100 M_m R_m^2} = \frac{16}{100} \approx \frac{1}{6}$$

or $\boxed{\frac{W_m}{W_e} = \frac{1}{6}}$

i.e., weight of the object on the Moon = $(1/6) \times$ its weight on the Earth

Now, $W_m = mg_m$ and $W_e = mg_e$, where g_m and g_e are accelerations due to gravity of Moon and Earth respectively.

$$\therefore \frac{W_m}{W_e} = \frac{mg_m}{mg_e} = \frac{1}{6}$$

or $\boxed{\frac{g_m}{g_e} = \frac{1}{6}}$



Numerical

3

Ability

1. What is the weight of a person whose mass is 50 kg ($g = 9.8 \text{ m/s}^2$)?

Solution

The weight of the person

$$W = mg = 50 \times 9.8 = 490 \text{ N}$$

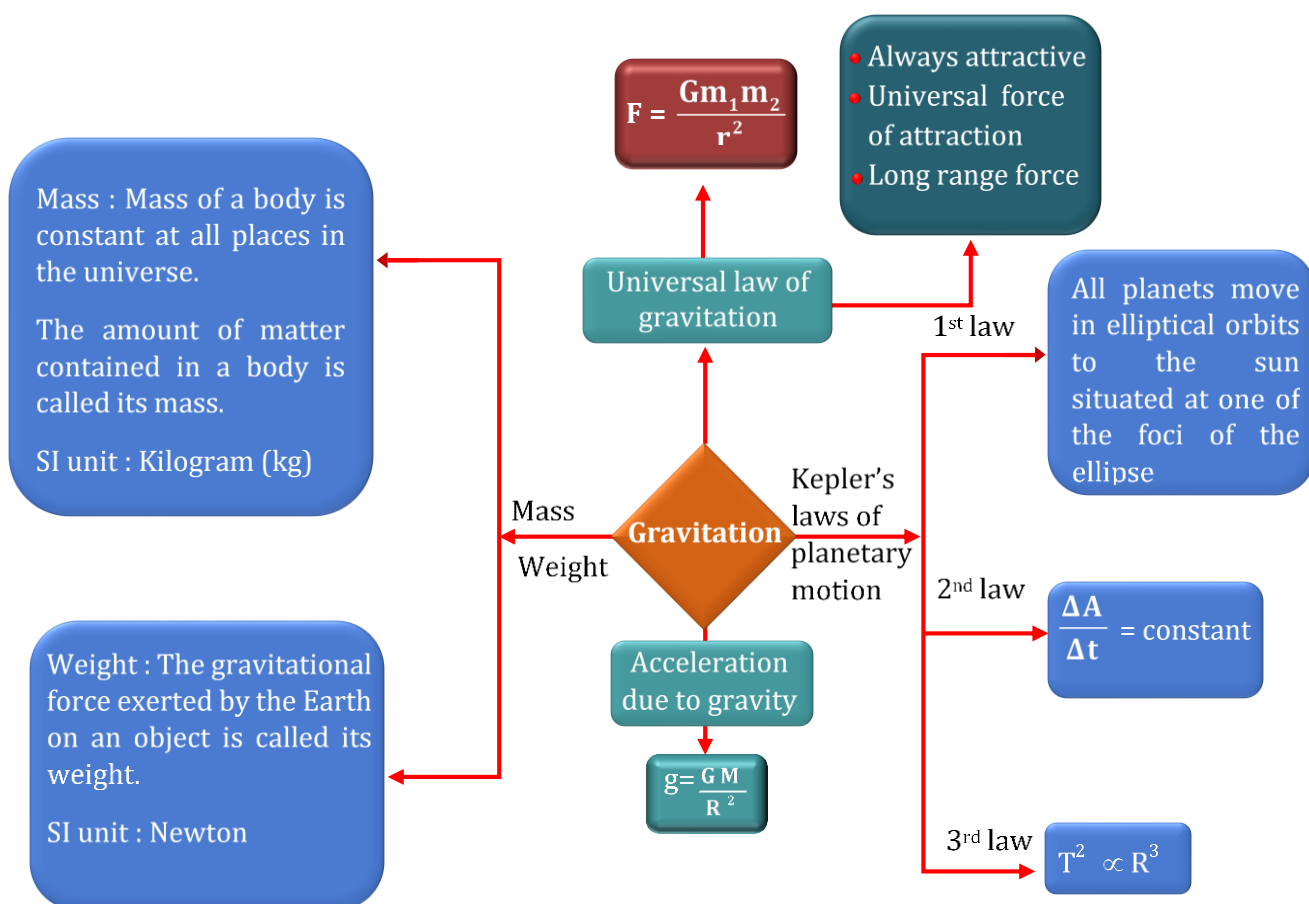
Decode the problem

Identify the equation

$$W = mg$$

(since, m and g are given)

Concept Map



Some Basic Terms

1. **Artificial Satellites** – An artificial satellite is an object that people have made and launched into orbit using rockets.
2. **Forecasting** – Forecasting refers to the practice of predicting what will happen in the future by taking into consideration events in the past and present.
3. **Interactions** – the action or influence of people, groups, or things on one another.
4. **Ordinary** – normal; not unusual or different from others.
5. **Galaxies** – A galaxy is a huge collection of gas, dust, and billions of stars and their solar systems.
6. **Significance** – the importance or meaning of something.
7. **Restricted** – controlled or limited.
8. **Physicists** – Physicists explore the fundamental properties and laws that govern space, time, energy, and matter.
9. **Orbit** – orbit is the curved trajectory of an object such as the trajectory of a planet around a star, or of a natural satellite around a planet, or of an artificial satellite around an object.
10. **Phenomena** – a fact or an event in nature or society, especially one that is not fully understood.
11. **Tides** – Tides are the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted by the Moon (and to a much lesser extent, the Sun) and are also caused by the Earth and Moon orbiting one another.
12. **Influence** – the power to affect, change or control somebody/something.
13. **Shrink** – to become smaller or make something smaller.
14. **Physical balance** – a physical balance is simply a weight measuring instrument. It is used to measure the weight of any unknown object with the help of the other object with known weight.
15. **Weighing machine** – a measuring instrument for weighing; shows amount of mass.
16. **Spring balance** – A spring balance is an instrument consisting of a hook attached to the end of a spring, used for weighing objects.
17. **Inherent** – that is a basic or permanent part of somebody/something and that cannot be removed.
18. **Exerts** – to make use of something, for example influence, strength, etc., to affect somebody/something.
19. **Apparent** – that seems to be real or true but may not be.
20. **Space shuttle** – a rocket-launched spacecraft able to land like an unpowered aircraft, used to make repeated journeys between the earth and space.

SOLVED EXAMPLES

1. What is the force of gravitation between two point masses of 1 kg and 2 kg kept 1 m apart?

Solution

$$m_1 = 1 \text{ kg}, m_2 = 2 \text{ kg}, r = 1 \text{ m}$$

$$F = G \frac{m_1 m_2}{r^2} = 6.67 \times 10^{-11} \times \frac{1 \times 2}{1^2}$$

$$= 13.34 \times 10^{-11} \text{ N.}$$

This is an extremely small force.

2. Calculate the force of gravitation due to a child of mass 25 kg on his fat mother of mass 75 kg if the distance between their centres is 1 m from each other. Given $G = (20/3) \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$.

Solution

$$\text{Here } m_1 = 25 \text{ kg}; m_2 = 75 \text{ kg}; d = 1 \text{ m};$$

$$G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$$

$$\text{Using } F = \frac{G m_1 m_2}{d^2}$$

$$\text{or } F = \frac{20 \times 10^{-11} \times 75 \times 25}{3 \times (1)^2}$$

$$\text{or } F = 12,500 \times 10^{-11}$$

$$\text{or } F = 1.25 \times 10^{-7} \text{ N}$$

3. The gravitational force between two objects is F . How will this force change when distance between them is reduced to half?

Solution

According to Newton's law of gravitation, gravitational force F between two objects distance r apart is

$$F \propto \frac{1}{r^2}$$

When distance between them is reduced to half, i.e., $r' = r/2$, the force, $F' \propto \frac{1}{r'^2}$

$$\text{Thus, } \frac{F'}{F} = \frac{r^2}{r'^2} = \frac{r^2}{(r/2)^2} = 4$$

$$\text{or } F' = 4F$$

i.e., force becomes 4 times its previous value.

4. The gravitational force between two objects is F . How will this force change when the mass of each object is quadrupled?

Solution

According to Newton's law of gravitation, the gravitational force F between two objects of masses m_1 and m_2 is $F \propto m_1 m_2$

When mass of each object is quadrupled,

$$m'_1 = 4m_1$$

$$\text{and } m'_2 = 4m_2$$

$$\text{The force, } F' \propto m'_1 m'_2$$

$$\text{Thus, } \frac{F'}{F} = \frac{m'_1 m'_2}{m_1 m_2} = \frac{(4m_1)(4m_2)}{m_1 m_2} = 16$$

$$\text{or } F' = 16 F$$

i.e., force becomes 16 times its previous value.

5. A ball reaches to the maximum height in 2 s. With what velocity was it thrown up?

Solution

$$\text{Here } u = ?; g = 9.8 \text{ m s}^{-2}; t = 2 \text{ s}; v = 0$$

$$\text{Using } v = u - gt$$

$$\text{or } 0 = u - 9.8 \times 2$$

$$\text{or } u = 19.6 \text{ m s}^{-1}$$

6. A ball was thrown up with a initial velocity of 19.6 m/s. How high did it go?

Solution

$$\text{Given, } u = 19.6 \text{ m/s}; g = 9.8 \text{ m s}^{-2}; v = 0$$

$$\text{Now, } v^2 = u^2 - 2gh$$

$$\text{or } 0^2 = (19.6)^2 - 2 \times 9.8 \times h$$

$$\text{or } h = 19.6 \text{ m}$$

7. A particle is thrown up vertically with a velocity of 50 m/s. (a) What will be its velocity at the highest point of its journey? (b) How high would the particle rise? (c) What time would it take to reach the highest point? ($g = 10 \text{ ms}^{-2}$)

Solution

(a) At the highest point the velocity will be zero.

(b) Considering activity A to B,

$$v^2 = u^2 - 2gh$$

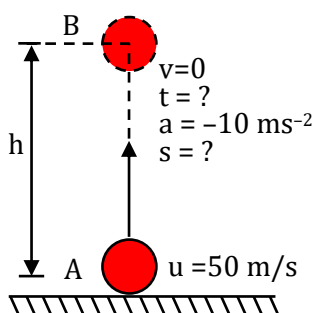
$$\text{or } 0^2 = (50)^2 - 2 \times 10 \times h$$

$$\text{or } h = \frac{50 \times 50}{2 \times 10} = \mathbf{125 \text{ m}}$$

(c) Using $v = u - gt$

$$\text{or } 0 = 50 - 10 \times t$$

$$\text{or } t = \frac{50}{10} = \mathbf{5 \text{ s}}$$



8. A particle is dropped from a tower 180 high. What is the velocity when it touches the ground? How long does it take to reach the ground?

Solution

Given, $u = 0$, $h = 180 \text{ m}$, $g = 10 \text{ m/s}^2$,

$v = ?$, $t = ?$

$$\text{from } v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2 \times 10 \times 180$$

$$v^2 = 3600 \text{ m/s}; v = 60 \text{ m/s}$$

Now, $v = u + gt$

$$60 = 0 + 10 \times t$$

$$60 = 10t \quad \text{or } t = \mathbf{6 \text{ s}}$$

9. A ball is thrown up with a speed of 15 m/s. How high will it go?

Solution

Given, $u = 15 \text{ m/s}$, $v = 0$, $g = 10 \text{ m/s}^2$, $h = ?$

$$\text{from } v^2 = u^2 - 2gh$$

$$\text{or } 0 = (15)^2 - 2 \times 10 \times h$$

$$\text{or } 15 \times 15 = 20h$$

$$\text{or } h = \frac{225}{20} = \mathbf{11.25 \text{ m.}}$$

10. An object is thrown vertically upwards and rises to height of 10 m. Calculate the velocity with which the object was thrown.

Solution

Given $h = 10 \text{ m}$, $v = 0$, $g = 9.8 \text{ m/s}^2$, $u = ?$

$$\text{from } v^2 = u^2 - 2gh$$

$$\text{or } 0 = u^2 - 2 \times 9.8 \times 10$$

$$\text{or } u^2 = 196$$

$$\text{or } u = \mathbf{14 \text{ m/s}}$$

11. A stone is released from the top of a tower of height 980 m. Calculate its velocity after 3 seconds.

Solution

Given, $u = 0$, $h = 980 \text{ m}$, $g = 9.8 \text{ m/s}^2$,

$t = 3 \text{ s}$, $v = ?$

$$\text{from } v = u + gt$$

$$v = 0 + 9.8 \times 3$$

$$v = \mathbf{29.4 \text{ m/s}}$$

12. A body is dropped freely under gravity from the top of tower of height 78.4 m. Calculate the time to reach the ground.

Solution

Given, $u = 0$, $h = 78.4 \text{ m}$, $g = 9.8 \text{ m/s}^2$

$$\text{from } h = ut + \frac{1}{2}gt^2$$

$$\text{or } 78.4 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2$$

$$\text{or } 78.4 = 4.9t^2$$

$$\text{or } t^2 = \frac{78.4}{4.9} = 16$$

$$\text{or } t = \sqrt{16} = \mathbf{4 \text{ sec.}}$$

13. Calculate the acceleration due to gravity on the surface of a planet whose mass is same as that of the earth and radius is twice that of earth.

Solution

Acceleration due to gravity is given by,

$$g = \frac{GM}{R^2}$$

For Earth, $g_E = \frac{GM}{R^2} = 9.8 \text{ m/s}^2$ (given)

Acceleration due to gravity for another

given planet, $g' = \frac{GM'}{(R')^2}$

Here, $M' = M$ and $R' = 2R$

$$\begin{aligned} \text{So, } g' &= \frac{G \times M}{(2R)^2} = \frac{GM}{4R^2} = \frac{g_E}{4} = \frac{9.8}{4} \\ &= 2.45 \text{ m/s}^2 \end{aligned}$$

14. What is the weight of a person whose mass is 50 kg ($g = 9.8 \text{ m/s}^2$) ?

Solution

The weight of the person

$$W = mg = 50 \times 9.8 = 490 \text{ N}$$

15. Weight of a girl is 294 N.

Find her mass.

$$(g = 9.8 \text{ ms}^{-2})$$

Solution

$$W = mg$$

$$294 = m \times 9.8$$

$$m = \frac{294}{9.8} = 30 \text{ kg}$$